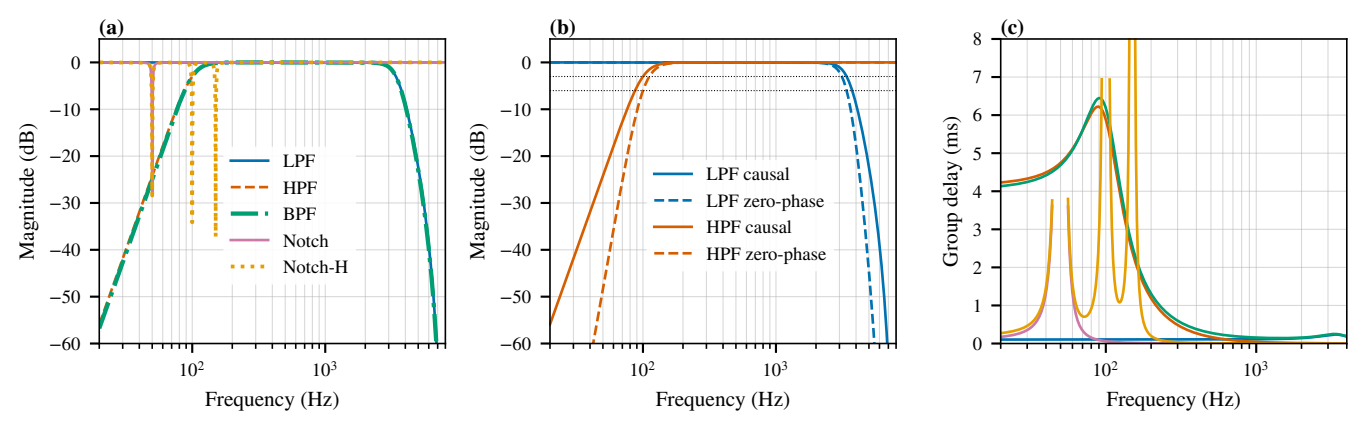
(a)
Magnitude (dB)
0
−10
−20
−30
−40
−50
−60
10^2
10^3
Frequency (Hz)
LPF
HPF
BPF
Notch
Notch-H
(b)
Magnitude (dB)
0
−10
−20
−30
−40
−50
−60
10^2
10^3
Frequency (Hz)
LPF causal
LPF zero-phase
HPF causal
HPF zero-phase
(c)
Group delay (ms)
8
7
6
5
4
3
2
1
0
10^2
10^3
Frequency (Hz)